

Comparison of two phenotypes to identify SNPs associated with poor reading

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Abstract

Reading disability and ability has a high heritability rate leading to speculation about the underlying genetic causes. Past research has examined the genetics for dyslexia and for reading ability, but past research has not often used a multi-factorial approach when considering reading ability. We compared the genetic associations for two phenotypes, reading disability (case-control) and a multifactorial reading ability phenotype, in the same sample. We found 4 SNPs in common, however after multiple test correction there were no SNPs in common. We have demonstrated how a multifactorial composite can extend our understanding of the genetics of reading disability and ability.

Introduction

- Common approaches for investigating the genes for reading disability and ability are case-control and single ability quantitative analyses
- Reading ability is based on many different skills, which suggests that using a multifactorial approach might be a better representation of ability, similar to educational attainment

Purpose: Compare GWAS results between two ways of representing reading disability and ability

Research Question 1: Are there SNPs in common between a case-control (reading disorder) phenotype and a multifactorial reading ability phenotype?

Research Question 2: What SNPs are associated with a multifactorial reading ability phenotype?

Methods

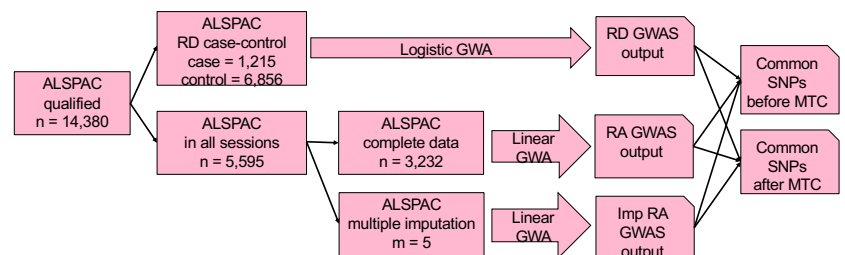
Data source. The Avon Longitudinal Study of Parents And Children (ALSPAC; Boyd et al., 2013) is a prospective birth cohort. Data collection began the early 1990s and is ongoing. The sample size at year 1 was 14,684. Behavioral data collection began at age 7 with focus sessions. A selection of the original participants plus new recruits were invited to attend these sessions.

Genotyping. ALSPAC children were genotyped using the Illumina HumanHap550 quad chip genotyping platforms by 23andme. After quality control (individual call rate > 0.97, SNP call rate > 0.95, Minor allele frequency (MAF) > 0.01, Hardy-Weinberg equilibrium (HWE) $P > 10^{-7}$, and removal of individuals with cryptic relatedness and non-European ancestry, 8,237 children and 477,482 SNP genotypes were retained. SNPs were flipped to the forward strand and haplotypes were estimated using ShapIT (v2.r644). Imputation was performed using Impute V2.2.2 in 1000 genomes reference haplotypes (Version 1 Phase 3). Minor changes with respect to previously reported sample numbers are due to withdrawn consents and alterations in GWAS quality control.

Participant selection. We excluded participants who had hearing loss, a diagnosis of ASD, and/or nonverbal IQ ≥ 72 . Additionally, we randomly selected one twin per pair for analysis. We also identified participants who had behavioral data for all Focus sessions and those with complete behavioral data. The number of participants varied depending on the phenotype of interest.

Methods

Phenotyping. Reading disorder status was defined as scoring less than -1 z-score on three out of ten reading measures. Reading ability is a continuous composite score. This composite score was based on a unifactor confirmatory factor model. Lambda values greater than 0.20 were extracted and used to calculate the composite score. A child's z-score on a measure was multiplied by the corresponding lambda value; these products were summed for all ten measures and the composite was scaled.

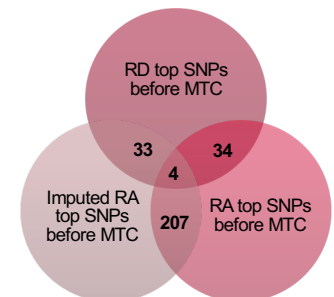


Results

	RD case	RD ctrl	Reading Ability	Imputed Reading Ability
Percent male	59%	47%	49%	49%
Nonverbal IQ	94.56	102.54	102.13	101.44
# SNPs			1	19
FDR $p < 0.05$	0			

Common significant SNP after multiple test correction

rs181384543 (chrX:22779030)
RA: B = -2.91 SE = 0.4996 MAF = 0.02
Imputed RA: B = -2.73 SE = 0.4903 MAF = 0.02



Conclusion

Research question 1: Although there are SNPs in common, these commonalities do not survive multiple test correction. This finding is limited because none of the top SNPs for our reading disorder phenotype survived multiple test correction.

Research question 2: We have evidence for one SNP being associated with our reading ability phenotype. This SNP is in an intronic region for *PTCHD1-AS* on chromosome X, which has previously been associated with autism spectrum disorder. *PTCHD1-AS* is non-coding RNA and is highly expressed in the brain, in particular the cerebellar and cerebellum.

Acknowledgements

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